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Brigit Muñoz

Taller de calculo de volumen de solidos de revolucion mediante integrales

Parte A - Metodo de disco.

1. $y = x^2$, $x = 0$, $x = 2$. Gira alrededor del eje x

Funcion $f(x) = x^2$

$R(x) = f(x) = x^2$

Al rededor del eje horizontal Pegada al eje de rotacion

$$V = \pi \int_a^b [R(x)]^2 dx$$

$$V = \pi \int_a^b ((x)^2)^2 dx$$

$$\pi \int_0^2 x^4 dx$$

$$V = \left| \frac{x^5}{5} \right|_0^2$$

$$V = \pi \left(\frac{2^5}{5} - \frac{0^5}{5} \right)$$

$$V = \pi \left(\frac{32}{5} - 0 \right)$$

$$V = \frac{32\pi}{5}$$

$$V = 20.106$$

2. $y = \sqrt{x}$, $x = 0$; $x = 4$

radio: $R(x) = f(x) = \sqrt{x}$

$$V = \pi \int_0^4 (\sqrt{x})^2 \, dx = \pi \int_0^4 x \, dx$$

$$V = \pi \left| \frac{x^2}{2} \right|_0^4$$

$$V = \pi \left(\frac{4^2}{2} - \frac{0^2}{2} \right)$$

$$V = \pi \left(\frac{16}{2} - 0 \right)$$

$$V = \pi (8)$$

$$V = 25.13$$

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Ponto 8

$x = y^2$, $0 \leq y \leq 3$. gire alrededor del eje x

radio = $R(y) - g(y) - y^2$

$$V = \pi \int_0^3 (y^2)^2 \, dy$$

$$V = \pi \int_0^3 y^4 \, dy$$

$$V = \pi \left(\frac{y^5}{5} \right) \Big|_0^3$$

$$V = \pi \left(\frac{3^5}{5} - \frac{0^5}{5} \right)$$

$$V = \frac{243}{5} - 0$$

$$V = 152.68$$

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D M A

Parte B. Metodo De la Arandela

6. Region entre $y = x + 2$ e $y = x$, para $0 \leq x \leq 3$

$$V = \pi \int_a^b [(R(x))^2 - r(x)^2]\, dx$$

$$V = \pi \int_0^3 [(x+2)^2 - (x)^2]\, dx$$

$$(x+2)^2 - x^2 = (x^2 + 4x + 4)$$

$$V = \pi \int_6^3 (4x + 4)\, dx$$

$$V = \pi \left| 4 \cdot \frac{x^2}{2} + 4x \right|_0^3$$

$$V = \pi \left| 2x^2 + 4x \right|_0^3$$

$$V = \pi \left| 2(3)^2 + 4(3) \right| 0 - (2(0)^2 + (4(0))$$

$$V = \pi (18 + 12) - 0$$

$$V = \pi (30)$$

$$V = 94.25$$

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7. Region entre $y = x^2+1$ & $y = 1$ para $0 \leq x \leq 2$

$$V = \pi \int_0^2 [(x^2+1)^2 - (1)^2]\, dx \qquad R(x) = x^2+1 \quad r(x) = 1$$

$$(x^2+1)^2 - 1 = (x^4 + 2x^2 + 1) =$$

$$V = \pi \int_0^2 (x^4 + 2x^2)\, dx$$

$$V = \pi \left(\frac{x^5}{5} + 2 \cdot \frac{x^3}{3} \right)\Big|_0^2$$

$$\pi \left(\frac{x^5}{5} + \frac{2x^3}{3} \right)\Big|_0^2$$

$$V = \pi \left(\frac{2^5}{5} + \frac{2(2)^3}{3} \right) - 0$$

$$V = \pi \left(\frac{32}{5} + \frac{16}{3} \right)$$

$$V = \pi \left(\frac{175}{15} \right)$$

$$V = 36 \cdot 86$$

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8. region entre $y=4$ e $y=x^2$ entre los puntos de intersección

límites $a=-2, b=-2$

$R(x)=4$

$r(x)=x^2$

$$V=\pi\int_{-2}^{2}[(4)^2-(x^2)^2]\,dx$$

$$V=\pi\int_{-2}^{2}(16-x^4)\,dx$$

$$V=\pi\left|16x-\frac{x^5}{5}\right|_{-2}^{2}$$

$x=2$

$$16(2)-\frac{2^5}{5}=32-\frac{32}{5}=\frac{128}{5}$$

$x=-2$

$$16(-2)-\frac{(-2)^5}{5}=-32+\frac{32}{5}=-\frac{128}{5}$$

$$V=\pi\left|\frac{128}{5}-\left(-\frac{128}{5}\right)\right|$$

$$=\pi\left(\frac{256}{5}\right)$$

$$=160.85$$

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